Roll number :.....

Amrita Vishwa Vidyapeetham

Midterm Examination October 2025

24MAT209: Transforms & PDE

S3 Int. MSc. Mathematics with minor in Data Science &

S3 Int. MSc. Physics with minor in Scientific Computing

Scientific calculators are allowed

Duration : 2 Hours

Maximum : 50 marks

Qn. No.	Questions	Marks	CO	Blooms Taxonomy Level
1	(a) State the general form of Fourier series for 2L-periodic functions. (b) Find the Fourier series of the function $f(x) = x - 3$, $-\pi < x < \pi$ and $f(x + 2\pi) = f(x)$, $x \in \mathbb{R}$.	10	CO1	L2
2	(a) Define the half-range sine series and cosine series for a function $f(x)$ defined on the interval $0 < x < L$. (b) Sketch $f(x) = e$, $0 < x < 2$ and its two periodic extensions. (c) Expand $f(x) = e$, $0 < x < 2$, in a half-range sine series.	10	CO1	L3
3	(a) Define orthogonal functions with respect to the weight function $r(x) > 0$, on the interval $[a, b]$. (b) Show that $sin(mx)$ and $sin(nx)$ are orthogonal functions with respect to the weight function $r(x) = 1$, on the interval $[-\pi, \pi]$ for $m \neq n$, $m, n \in \mathbb{Z}$.	10	CO2	L2
4	(a) State the general form of the Sturm-Liouville problem on an interval $[a, b]$. (b) Solve the Sturm-Liouville problem $y'' + \lambda y = 0$, $0 < x < 1$, with boundary conditions $y(0) = 0$, $y(1) = 0$. Find the eigenvalues and eigenfunctions.	10	CO2	L3
5	(a) Define Fourier integral for a piecewise continuous function $f(x)$ for $x \in \mathbb{R}$. (b) Find the Fourier integral of $f(x) = \begin{cases} x + \pi, & \lvert x\rvert \leq 3. \\ 0, & \lvert x\rvert > 3. \end{cases}$	10	CO3	L2

L1-Remember, L2-Understand, L3-Apply, L4-Analyze, L5-Evaluate, L6-Create